

Practice Set – Unit-3

1. Find power set of $X = \{\phi, 1, \{1,2\}, \{1,2,3\}\}$
2. Use Venn diagrams to show that $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$ & $(A \cup B \cup C)' = A' \cap B' \cap C'$
3. List the ordered pairs in the relation R from $A = \{1,2,3,4,5,6,7,8,9,10\}$ to $B = \{5,6,7,8,9,10,11,12\}$, where $(a,b) \in R$ if and only if
 - (a) $b-a = 5$
 - (b) a divides b
 - (c) $\gcd(a,b) = 2$
4. Give an example of a relation which is:
 - (a) reflexive but neither symmetric nor transitive;
 - (b) symmetric but neither reflexive nor transitive;
 - (c) transitive but neither reflexive nor symmetric;
 - (d) reflexive and symmetric but not transitive;
 - (e) reflexive and transitive but not symmetric;
 - (f) symmetric and transitive but not reflexive.
5. Show that the relation R defined on the set $\mathbb{R}$ of real numbers as
$$R = \{(a, b) : a \leq b^2\}$$
is neither reflexive nor symmetric nor transitive.
6. Let $S = \{1,2,3,4,5,6,7,8,9,10\}$. Determine whether each of the following is a partition of S . Explain.
 - (a) $P_1 = \{\{1,2,3,7\}, \{1,4,5,6\}, \{8,9,10\}\}$;
 - (b) $P_2 = \{\{1,2\}, \{3,5,6\}, \{7,8,9,10\}\}$;
 - (c) $P_3 = \{\{1,3,5,7,9\}, \{2,4\}, \{6\}, \{8,10\}\}$.
7. Let R be the following equivalence relation on the set $A = \{a,b,c,d,e,f\}$:
$$R = \{(a,a), (a,b), (b,a), (b,b), (c,c), (c,e), (c,f), (d,d), (e,c), (e,e), (e,f), (f,c), (f,e), (f,f)\}.$$
Determine the equivalence classes of A and thus find the partition of A using equivalence classes of R .
8. Let R be the relation defined on the set $\mathbb{Z}$ of integers as follows: for integers a and b ,
$$aRb \text{ if and only if } a \equiv b \pmod{3},$$
that is, aRb if 3 divides $(a-b)$.
 - (a) Show that R is an equivalence relation on $\mathbb{Z}$.
 - (b) What are the equivalence classes of R ?
9. How many relations are there on a set with n elements? How many reflexive relations are there on a set with n elements?

10. Let $R = \{(a,d),(a,e),(b,c),(b,d),(c,a),(c,c),(d,e)\}$ and $S = \{(b,a),(c,a),(d,b),(e,c)\}$ be relations defined on $\{a,b,c,d,e\}$.
- Find $S \circ R$ and $R \circ S$.
 - Find R^{-1} and S^{-1} .
 - Show that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.
11. Let $A = \{1,2,3,4,5\}$, $B = \{1,2,3,\dots,100\}$. Determine which of the following relations from A to B are functions:
- $R = \{(1,1), (2,1), (3,1), (4,1), (5,1)\}$
 - $R = \{(1,1), (1,2), (1,3), (1,4), (1,5)\}$
 - $(a,b) \in R$ if $b = a^2$.
 - $(a,b) \in R$ if $a^2 = b$.
12. In each of the following cases, state whether the function is one-one, onto, or bijective. Justify your answer.
- $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$;
 - $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1 + x^2$.
13. In the above question, determine the composition functions $f \circ g$ and $g \circ f$. Moreover, in case the function is bijective, find its inverse.
14. Consider functions f and g such that composite $g \circ f$ is defined and is one-one. Are f and g both necessarily one-one? Are f and g both necessarily onto, if $g \circ f$ is onto?
15. If $f(x) = \frac{4x+3}{6x+4}$, $x \neq \frac{2}{3}$, show that $f \circ f(x) = x$ for all $x \neq \frac{2}{3}$. What is the inverse of f ?
16. Show that the set of even integers is countable.
17. Show that the set of real numbers is an uncountable set.
18. Use the Principle of Mathematical Induction to show that $4n - 1$ is divisible by 3, for each natural number n .
19. Prove that number of subsets of a set containing n distinct elements is 2^n , for all $n \in \mathbb{N}$.
20. Show that $n^2 < 2^n$ for all natural numbers $n \geq 5$.
21. Use Principle of Strong Induction to show that if n is an integer greater than 1, then n can be written as the product of primes.
22. Express each number as a product of its prime factors:
- 140

- (b) 3825
- (c) 5005
- (d) 7429

23. Use the fundamental theorem of arithmetic to show that the numbers 4^n , where n is a natural number, do not end with the digit zero.

24. Use Euclid's algorithm to find the greatest common divisor:

- (a) 64 and 168
- (b) 42 and 234
- (c) 1001 and 1441